

Data Ethics Case Study: Historical Bias in Automated Recruitment

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1. Data Ethics Challenges Encountered/Identified

This case study examines two central data ethics issues:

* Historical Bias in Training Sets (Deon Item 1.1 - Representation): When models train on historical data, they learn to reproduce past societal biases and inequalities rather than objectively measuring present applicant merit.

* Proxy Variable Discrimination (Deon Item 3.2 - Fairness & Auditability): Simply removing explicit demographic fields (such as gender or race) does not make a model neutral. Algorithms automatically identify correlated "proxy" features-such as specific phrasing, school names, or activities that indirectly encode protected attributes.

2. Real-World Case Analysis: Automated Resume Screening

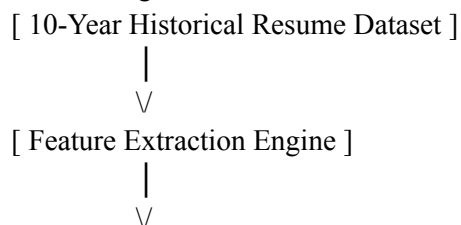
Problem Overview

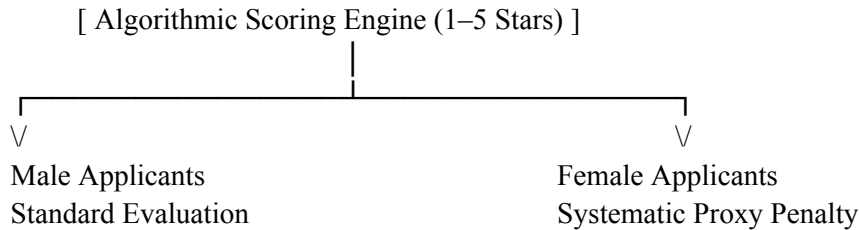
In 2018, details and information showed up that Amazon abandoned on internal automated hiring engine developed in 2014. The software scored software developers and other STEM candidates on a 1 to 5 star score scale to moderate recruitment.

Operational Mechanism

The machine model trained on resumes submitted to the company over a 10-year operation period. Because the technology sector had historically been dominated by males during this period, the training dataset consisted of large amount of male applicant profiles. This had a large bias.

Problem Diagram:





The system learned that male and their attributes correlated highly with higher hiring rate. When researchers studied this and try to remove the obvious gender variables, the model got used to the environment by penalizing resumes that contained proxy indicators, like the word `"women's"` (e.g., `"women's chess club captain"`) or graduates of all-women colleges.

3. Author Analysis and Mitigation Framework

Personal Reflection and Academic Citations

Removing all features such as gender or anything relative to gender doesn't make the model fair and can lead to a skew, because algorithms are built to find hidden patterns. Those gender features are necessary to make additional statistical pattern recognition. When obvious demographic fields are deleted, the model uses other clues like extracurriculars or education as proxy/substitute variable to replace the missing information.

Traditional software engineering and data science process errors cause immediate code breakdowns that are easy to spot because of the compiler, but historical dataset bias is completely invincible to that standard error log and it just automates with the remaining data (including human discrimination categories).

Because of these hidden factors and risks, machine learning engineers and technical statisticians must execute through data audits, analysis, and fairness checks rather than treating algorithm as a default tool. AI systems should be treated as a back-up assistant rather than a go-to decision maker.

Technical Safeguards and Applicable Recommendations

- * **Pre-Training EDA:** Perform demographic audits on historical and past datasets to research representation gaps before selecting a appropriate model.

- * **Counterfactual Testing:** Testing models by swapping proxy features like club or university names in test samples to ensure predictions remain common and equal.

- * **Human-in-the-Loop:** Humans should always fact check the AI, need to learn to use automated tools only as an advisory tool rather than it having full control as an autonomous rejection algorithm.

4. References and Academic Citations

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3. Industry Ethics Standard:

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